

Tribhuvan University
Institute of Science and Technology
2069

Bachelor Level/ First Year/ First Semester/ Science
Computer Science and Information Technology (MTH:104)
(Calculus and analytical Geometry)

Full Marks: 80
Pass Marks: 32
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all questions.

Group A (10×2=20)

1. Verify the mean value theorem for the function $f(x) = \sqrt{x(x-1)}$ in the interval $[0, 1]$.
2. Find the length of the curve $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1$ for $0 \leq x \leq 1$.
3. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n!}$ By comparison test.
4. Obtain the semi-major axis, semi-minor axis, foci, vertices $\frac{x^2}{25} + \frac{y^2}{16} = 1$.
5. Find the angle between the vectors $2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $-4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.
6. Obtain the area of the region R bounded by $y=x$ and $y=x^2$ in the first quadrant.
7. Show that the function

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Is continuous at every point in the plane except the origin.

8. Using partial derivatives, find $\frac{dy}{dx}$ if $2xy + \tan y - 4y^2 = 0$.
9. Verify that the partial differential equation $\frac{d^2z}{dx^2} - \frac{d^2z}{dy^2} = \frac{2z}{x}$ is satisfied by

$$z = \frac{1}{x} \phi(y-x) + \phi'(y-x)$$

10. Find the general solution of the equation

$$x^2 \frac{dz}{dx} + \frac{y^2 dz}{dy} = (x, y)z.$$

Group B (5×4=20)

11. State and prove mean value theorem for definite integral.
12. Find the area of the region that lies in the plane enclosed by the cardioid $r = 2(i + \cos\theta)$.
13. What do you mean by principle unit normal vector? Find unit tangent vector and principle unit vector for the circular motion $\vec{r}(t) = (\cos 2t)\mathbf{i} + (\sin 2t)\mathbf{j}$.

- 14.** Define partial derivative of a function $f(x,y)$ with respect to x at the point (x_0, y_0) . State Euler's theorem, verify it for the function $f(x,y) = x^2 + 5xy + \sin x + 7e^x$,
 $x = \left(\frac{y}{2}\right) + 1$

- 15.** Find the particular integral of the equation $\frac{d^2z}{dz^2} - \frac{dz}{dy} = 2y - x^2$

Group C (5×8=40)

- 16.** Graph the function $y = x^{\frac{1}{2}} - 5x^{\frac{2}{3}}$
- 17.** What is meant by Maclaurin series? Obtain the Maclaurin series for the function $f(x) = e^{-x}$.
- 18.** Evaluate the double integral $\int_0^4 \int_{x=\frac{y}{2}}^{x=\frac{y}{2}+1} \frac{2x-y}{2} dx dy$ by applying the transformation $u = \frac{2x-y}{2}, v = \frac{y}{2}$ and integrating over an appropriate region in the uv -plane.
- 19.** Define maximum and minimum of a function at a point. Find the local maximum and local minimum of the function $f(x,y) = 2xy - 5x^2 - 2y^2 + 4x + 4y - 4$.
- 20.** Find the solution of the equation $\frac{d^2z}{dx^2} - \frac{d^2z}{dy^2} = x - y$

Or

Find the particular integral of the equation

$$(D^2 - D)z = 2y - x^2 \quad \text{Where } D = \frac{d}{dx}, D' = \frac{d}{dy}$$